
Supplementary Material for “BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach”

A PROOFS AND ADDITIONAL ANALYSIS

A.1 Behavior anchor and Proposition 1

Let $b(s) = \mathbb{E}[a \mid s]$. The conditional bias-variance identity

$$\mathbb{E}[\|\mu(s) - a\|_2^2 \mid s] = \|\mu(s) - b(s)\|_2^2 + \mathbb{E}[\|a - b(s)\|_2^2 \mid s]$$

proves Eq. (7) after averaging over states and dividing by d ; its second term is independent of μ . For deterministic policies, $W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2$, so d_d^2 is a fixed rescaling of integrated squared Wasserstein distance. Dividing the normalized anchored objective $-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$ by α gives Eq. (8) with $h = \alpha/2$. More generally, $-\lambda \mathbb{E}_s \hat{Q}$ gives $h = \lambda C/2$ because $\hat{Q} = C\bar{Q}$.

A.2 Proposition 2 and critic-error transfer

By ε_k -optimality relative to the feasible comparator μ_{k-1} ,

$$\begin{aligned} & -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_k(s)) + d_d^2(\mu_k, \mu_{k-1}) \\ & \leq -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_{k-1}(s)) + \varepsilon_k. \end{aligned} \tag{1}$$

Rearranging yields Eq. (13). For $\hat{Q} = Q^\dagger + e$, substitution further gives

$$\begin{aligned} m_k^{\hat{Q}} &= \frac{2h_k}{C_k} \mathbb{E}_s [Q^\dagger(\mu_k) - Q^\dagger(\mu_{k-1})] - d_d^2(\mu_k, \mu_{k-1}) \\ & \quad + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})] \\ &= m_k^{Q^\dagger} + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})]. \end{aligned} \tag{2}$$

This proves Eq. (14). If $|e(s, a)| \leq \delta_k$ on the realized path, then

$$m_k^{Q^\dagger} \geq -\varepsilon_k - \frac{4h_k}{C_k} \delta_k. \tag{3}$$

The result concerns a trusted critic objective. A true-return statement would require additional occupancy and value-estimation assumptions.

A.3 Proof of Proposition 3

Let $F_{s'}(a) = \min_j Q_j^-(s', a)$. If every $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz, then

$$\begin{aligned} |F_{s'}(a) - F_{s'}(b)| &\leq \max_j |Q_j^-(s', a) - Q_j^-(s', b)| \\ &\leq L(s') \|a - b\|_2. \end{aligned} \tag{4}$$

Therefore,

$$|y_{\mu_1} - y_{\mu_0}| \leq \gamma L(s') \|\mu_1(s') - \mu_0(s')\|_2. \tag{5}$$

Squaring, averaging over dataset transitions, and taking the square root proves Eq. (15). Later actors do not occur because the TD target conditions only on μ_1 .

A.4 Local explicit–implicit equivalence

Let $\beta = dh/C$, $g = \nabla_a \hat{Q}(a_0)$, and suppose the action constraint is inactive. The action-metric-matched linearized update is $\Delta a_{\text{lin}} = \beta g$. A statewise exact proximal solution satisfies

$$\Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (6)$$

If the Hessian of $\hat{Q}$ is locally Lipschitz near a_0 ,

$$\nabla_a \hat{Q}(a_0 + \Delta a) = g + H\Delta a + O(\|\Delta a\|^2). \quad (7)$$

Thus

$$(I - \beta H)\Delta a_{\text{prox}} = \beta g + O(\beta \|\Delta a_{\text{prox}}\|^2). \quad (8)$$

For $\beta \|H\| < 1$ and sufficiently small β ,

$$\Delta a_{\text{prox}} = \beta g + \beta^2 Hg + O(\beta^3), \quad (9)$$

whereas $\Delta a_{\text{lin}} = \beta g$. With only C^2 regularity, the safe remainder is $o(\beta^2)$ rather than $O(\beta^3)$. Both ideal updates are first-order schemes for fixed positive C , and changing a fixed C only rescales time. A positive policy-dependent global C preserves the raw-critic ascent direction under an adaptive time parameterization in the infinitesimal limit. This derivation is not applied to the large-budget regime with active projection, ReLU-region crossings, critic drift, or inaccurate actor solves.

A.5 Proof of Proposition 4

Write $x_j = x_0 + jhf_0 + O(h^2)$ for fixed K , which follows inductively from the one-step consistency assumption. Smoothness then gives

$$f(x_j) = f_0 + jhJ_0f_0 + O(h^2). \quad (10)$$

Summing the K increments,

$$\begin{aligned} x_K - x_0 &= h \sum_{j=0}^{K-1} f(x_j) + h^2 \sum_{j=0}^{K-1} p(x_j) + O(Kh^3) \\ &= Khf_0 + h^2 \frac{K(K-1)}{2} J_0f_0 \\ &\quad + Kh^2 p_0 + O(h^3). \end{aligned} \quad (11)$$

where the final remainder uses fixed K . Substituting $h = T/K$ proves Eq. (18). The exact flow satisfies $x(T) = x_0 + Tf_0 + \frac{1}{2}T^2J_0f_0 + O(T^3)$. Thus explicit Euler has leading error $-T^2J_0f_0/(2K)$, while backward Euler has $+T^2J_0f_0/(2K)$. Both remain first-order methods.

B EXPERIMENTAL DETAILS

B.1 Algorithms and schedules

The implementation parameter is $T = \tau = \alpha/2$, and the normalized budget is split as $h = T/K$. Thus $K = 1$ recovers TD3+BC with coefficient $\alpha = 2T$. BAR-Prox uses Eq. (9) with a moving reference; BAR-Lin constructs Eq. (19) and uses MSE regression. The K actor networks are independently initialized persistent networks with separate Adam states. Each iteration first updates the online critic. On every second iteration, one optimizer step is then performed per actor in chain order on the same minibatch: actor one updates first; its target actor and the updated critic each receive a Polyak update of .005; later actors then use stop-gradient references from the updated online chain. Equations (5) and the main-text linearized scale define the actual normalization, with safeguard 10^{-6} . All methods share data, critic architecture, and 10^6 critic updates, while a K -hop run performs K actor optimizer steps and is not compute matched.

The ideal proximal map is not a certificate for this live chain. For a later persistent actor that differs from its predecessor, even $h = 0$ leaves

$$\nabla_{\theta_k} \mathbb{E}_s [\|\mu_k(s) - \text{sg } \mu_{k-1}(s)\|_2^2/d] \neq 0 \quad (12)$$

in general. One Adam step therefore need not become a near-identity map as the nominal coefficient vanishes; the ideal analysis supplies local geometry, while the sweep evaluates the implemented optimizer trajectory.

We use a 256–256 ReLU actor MLP with tanh output and twin 256–256 ReLU critic MLPs, batch size 256, Adam with learning rate 3×10^{-4} , discount .99, policy delay 2, target noise .2 and clip .5 times the action bound, and state normalization by dataset mean and standard deviation plus 10^{-3} . Every reported sweep score is the ten-episode D4RL normalized mean at the final 10^6 -update checkpoint. Archived configurations record 50,000-update intermediate evaluation for K1/P2/P3, mixed 50,000/final-only evaluation for L2, and final-only evaluation for L3; exact K4 run configurations were not retained. Intermediate cadence does not enter the headline estimand. The 14-value grid spans .05 to 20. Proximal procedures use seeds 0–3, while linearized procedures use seeds 0–1.

The nine exact datasets are the Cartesian product of domains **halfcheetah**, **hopper**, and **walker2d** with suffixes **medium-v2**, **medium-replay-v2**, and **expert-v2**. Timeout rows are omitted from transition construction; terminal flags stop bootstrapping.

B.2 Diagnostic quantities

On a fixed current-state validation batch with paired dataset actions, we compute the empirical action-averaged proxies

$$\begin{aligned}\hat{D}_{\text{critic}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_1(s) - a\|_2^2, \\ \hat{D}_{\text{final}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_K(s) - a\|_2^2.\end{aligned}$$

These quantities measure the online actor chain relative to its sample anchor. They are proxies for, rather than direct measurements of, the next-state target-policy displacement in Proposition 3: the practical TD target uses a Polyak target actor and policy-smoothing noise. By Eq. (7), the population expectation of each proxy equals squared displacement from the conditional-mean behavior map plus the same additive conditional-action variance. Population differences therefore cancel this common floor, but ratios do not: a ratio of sample-anchored proxies is not a ratio of mean-policy displacements. Finite validation batches add sampling error to both interpretations. We additionally evaluate the deterministic Polyak target actor and 64 draws of clipped target-smoothing noise on 4096 next-state transitions per checkpoint. For source index i , the audit pairs $s'_i = s_{i+1}$ with dataset action a_{i+1} ; it excludes timeout rows and excludes true terminal rows from this bootstrap-only sample. All target-exposure quantities remain squared displacement per action coordinate relative to the paired next dataset action. Thus absolute levels include conditional action variance, whereas paired differences use a common sampled anchor. For BAR-Lin, projection attenuation is the RMS norm ratio between the clipped and raw target displacements. If $\tilde{g}$ is the normalized critic action gradient and Δa is the learned actor displacement, define

$$h_{\text{eff}} := \frac{\mathbb{E} \langle \Delta a, \tilde{g} \rangle}{\mathbb{E} \|\tilde{g}\|_2^2}.$$

We report $h_{\text{eff}}/(dT/K)$ as the effective fraction of the nominal linear-target coefficient. The matched audit evaluates 512 fixed states per environment–seed run. We also record action saturation, TD-error quantiles, critic magnitude, critic loss, and gains under both the jointly trained critic and a controlled common critic: the frozen final TD3+BC $T = 1$ critic from a separate canonical reference run outside the fourteen-value sweep, matched by environment and seed. A run-total gain is the sum of mean Q changes across the K hops on this shared batch. The common critic supports cross-critic comparison but is not treated as ground truth. A run is descriptively labeled collapsed only from its external normalized return, using the threshold 20.

For the exploratory simulator-reference audit, we clone MuJoCo integration state and environment RNG from evaluation-policy trajectories, record the episode clock, and reset the TimeLimit for a continuing-task critic-calibration estimand. At each state we compare $a_1 = \mu_1(s)$ and $a_K = \mu_K(s)$ under continuations ρ_1 , μ_1 , and μ_K , where ρ_1 is the Polyak first target actor with TD3 clipped target-smoothing noise. The primary calibration target is the discounted return under this effective backup continuation. The screening covers three environments, $T \in \{4, 7, 10, 14, 20\}$, $K \in \{2, 3\}$, two seeds, and 12 states per checkpoint. It uses a 1000-step horizon, $\gamma = .99$, four common-random-number target-smoothing rollouts, and a one-return-unit ranking dead zone. It is a post-hoc screening study, not a causal routing intervention.

Table 1: Mean score over nine tasks at each total budget. Proximal rows use four seeds; linearized rows use two.

T	.05	.1	.2	.4	.7	1.5	2.5	4	7	10	12	14	17	20
TD3+BC	63.81	62.57	64.58	65.94	67.69	74.79	67.41	45.93	30.36	23.77	21.64	18.96	18.92	18.09
Prox-2	60.45	64.19	67.90	69.61	73.65	78.05	80.66	74.48	58.88	41.14	32.17	29.55	25.95	26.25
Prox-3	60.14	60.82	68.08	71.01	73.32	78.95	81.41	81.26	71.00	51.14	49.74	41.68	32.75	35.83
Prox-4	59.59	63.95	68.71	71.26	73.26	77.70	81.25	82.06	81.79	74.35	67.88	54.28	52.14	35.29
Lin-2	60.98	63.12	66.20	72.07	67.98	72.70	80.10	73.95	45.63	39.98	27.09	25.10	24.41	20.63
Lin-3	58.32	59.97	63.69	70.52	73.23	77.14	76.94	75.31	66.47	45.78	30.65	39.54	25.74	23.88

The route-only intervention trains two actor-detached shadow critics from identical initial parameters on the same minibatches and target-smoothing draws while the primary first-hop critic continues to drive the shared actor trajectory. The four environment–budget cells—Hopper-medium at $T \in \{10, 20\}$ and Walker2d-medium at $T \in \{4, 7\}$, each with seeds 0–1—were selected post hoc from visible stability-frontier transitions, so inference is descriptive of this targeted sample. One shadow uses the first-hop target actor and the other the final-hop target actor. The simulator audit evaluates 12 cloned states at four checkpoints in each of eight runs; inference aggregates checkpoint-level state medians to the training-run level.

The compute-matched study uses the same eight final cells. Sequential $K = 3$ performs actor updates anchored at μ_0, μ_1 , and μ_2 ; the fixed-reference control anchors both later actors at μ_1 ; the direct-final control applies two optimizer calls to one $2T/3$ objective anchored at μ_1 . Every condition performs three actor optimizer calls per delayed update. The target-lag control instead keeps $K = 1$ and changes only the actor-target Polyak rate from .005 to .005/3, leaving the critic-target rate unchanged.

The frozen-critic small-step audit uses the final TD3+BC $T = 1$ critic for all nine environments and two seeds. At 512 interior dataset actions per run it compares the projected explicit step with a projected fixed-point solution of the implicit action equation for $h \in \{.001, .003, .01, .03, .1\}$. This is an action-level local-consistency test, not a return or coupled-training experiment.

B.3 Estimands and uncertainty

The primary high-budget estimand uses the upper seven values of the ordered fourteen-point grid, $T \geq 4$. Within each task it averages the available seeds and those seven budgets, then weights the nine task means equally. This is an operational stress-region aggregate, not a claim of pointwise monotonicity in T or task. Moving the boundary to $T \geq 10$ preserves the strict proximal ordering: 20.27, 31.01, 42.23, 56.79 for $K = 1, 2, 3, 4$. A seed-mean environment–budget cell is called collapsed when its normalized score is below 20. Thresholds 0, 10, 20, 30, and 40 expose sensitivity to that descriptive boundary.

For score contrasts we resample the nine task-level high-budget means with replacement for 100,000 draws using analysis seed 20260830. Raw-run collapse risk is first averaged within each task and then resampled by task using the same draw count and seed. These percentile intervals preserve the complete within-task budget and seed structure but do not remove dependence among the three datasets sharing one dynamics family. Cross- K means estimate each randomized procedure’s marginal performance. Identical critic initialization across depth is not required for validity; common random numbers would reduce variance. Paired rescue counts remain descriptive under the realized seed coupling.

C ADDITIONAL RESULTS

Across the four-seed proximal grid, P4 minus TD3+BC gains 38.59 points with task-resampling interval [21.39, 54.63] and eight of nine positive task means. At the score-20 threshold, proximal $K = 3 \rightarrow 4$ rescues eight high-budget cells and reverses none. Those cells contribute 54.7% of the total P4-minus-P3 score gain summed over all 63 high-budget cells. The mean gain is 12.06 with task-resampling interval [3.84, 21.11]; the raw collapse-risk difference is $-.119$ with task-resampling interval $[-.218, -.024]$. The two-seed linearized $K = 2 \rightarrow 3$ transition rescues seven cells and reverses two. These thresholded transitions are descriptive summaries, while the continuous score contrasts retain their magnitude.

Both disjoint seed pairs reproduce the strict $K = 1 < 2 < 3 < 4$ high-budget aggregate ordering. On seeds 2–3,

Table 2: Collapse-count sensitivity for $T \geq 4$. Entries are seed-mean cells / raw seeded runs. Cell denominators are 63; raw denominators are 252 for proximal and 126 for linearized rows.

Threshold	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
0	1/20	0/14	0/13	0/10	0/6	2/12
10	30/134	22/103	15/81	6/50	28/54	21/46
20	34/142	22/111	16/84	8/54	29/58	24/53
30	36/149	26/115	18/88	11/55	29/63	24/55
40	38/165	30/120	19/96	14/58	32/64	24/57

Table 3: Per-environment high-budget ($T \geq 4$) means.

Environment	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
Hopper-M	11.1	28.7	63.7	84.7	25.5	46.3
Hopper-MR	45.6	92.3	99.6	99.6	90.5	99.4
Hopper-E	3.1	16.8	28.2	46.4	16.6	27.7
HalfCheetah-M	58.8	58.9	58.9	58.2	53.4	55.8
HalfCheetah-MR	51.0	51.8	51.8	52.6	46.5	49.2
HalfCheetah-E	19.9	23.3	30.5	32.8	18.8	22.4
Walker2d-M	9.3	26.0	37.2	59.8	9.3	26.5
Walker2d-MR	27.3	51.2	72.4	76.8	51.2	43.5
Walker2d-E	2.2	21.8	25.1	65.1	18.5	24.4

P4 minus TD3+BC gains 39.54 points with a task-resampling interval of [21.54, 56.11] and eight of nine positive task means, closely matching the 37.64 gain, [20.51, 54.35] interval, and eight-of-nine direction on seeds 0–1. Raw collapse counts for $K = 1, 2, 3, 4$ are 68, 53, 41, 29 on seeds 0–1 and 74, 58, 43, 25 on seeds 2–3. The pairs share one method family and are not independent implementations.

The paired route-only study uses the same eight runs and aggregates four checkpoints per run. For route $r \in \{\text{first}, \text{final}\}$, its statewise symmetric relative error against the common first-route continuation is

$$e_r(s) = \frac{|Q_r(s, a_K) - Q_{\text{MC}}^{\rho_1}(s, a_K)|}{|Q_r(s, a_K)| + |Q_{\text{MC}}^{\rho_1}(s, a_K)| + 10^{-12}}, \quad \Delta(s) = e_{\text{final}}(s) - e_{\text{first}}(s). \quad (13)$$

The primary statistic takes the state median of Δ within each checkpoint and then averages the four checkpoints within a run. All eight run-level checkpoint-aggregate differences are positive. The direction holds in only five of eight comparisons when restricted to the final checkpoint. Under the primary aggregation, the mean is .124, median is .013, exact two-sided sign-flip $p = .0078$, and run-bootstrap mean 95% interval is [.008, .323]. Because the routes imply different continuation policies, this estimates relative alignment with the primary first-route value estimand rather than own-continuation calibration; it is not a return-effect estimate.

In the frozen-critic audit, 15 of 18 environment–seed runs have an eligible converged, projection-free point. Every eligible run has maximum local relative explicit–implicit discrepancy below 2.7%, with median run maximum .55%. The remaining three have no eligible point. This supports local numerical agreement, not a universal convergence rate.

C.1 Direct target exposure and simulator-reference screening

The direct target-action audit covers 270 method checkpoints: TD3+BC, P2, and P3 over nine environments, five budgets, and two seeds. It directly tests the P3 exposure signature and does not measure P4 target movement. Table 8 separates critic-target exposure from final-policy reach. The ratios are descriptive ratios of sample-anchored squared displacements and are not ratios of conditional-mean policy distances.

For the final-policy proxy, the P3-minus-TD3 arithmetic mean difference is $-.104$ with task–seed-cluster interval $[-.194, -.026]$, even though the median P3/TD3 ratio is 1.057. A few large contractions dominate the mean, whereas the typical cell expands slightly.

The current-state proxy and deterministic target exposure have Pearson correlation .9999 over all 270 cells; their median ratio is .997 and the largest cellwise relative difference is 5.7%. The corresponding correlation after

Table 4: Core matched $K = 2$ diagnostic medians over $T \in \{4, 7, 10, 14, 20\}$. Stable and collapsed groups are defined by external return, while the remaining rows diagnose the learned trajectory.

	Lin st.	Lin col.	Prox st.	Prox col.
No. runs	52	38	59	31
TD p99	270	9.6×10^{23}	160.8	5.0×10^{21}
$\hat{D}_{\text{critic}}$	.112	.538	.064	.446
$\hat{D}_{\text{final}}$	.147	.626	.094	.578
Common- Q gain	+2.11	-1.42	+2.51	-1.06

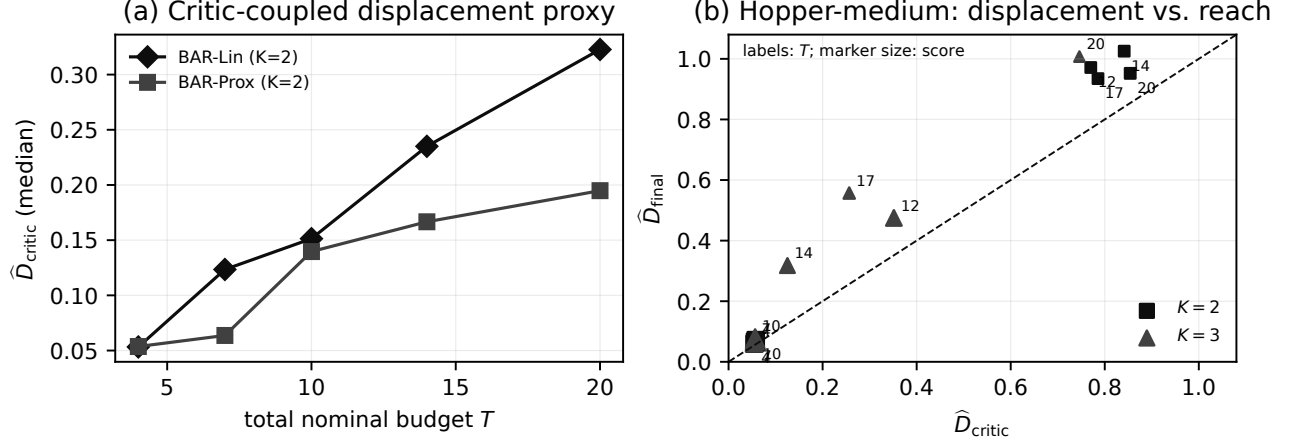


Figure 1: Realized movement. (a) The proximal realization produces a smaller current-state first-hop displacement proxy for $T \geq 7$. (b) On Hopper-medium, $K = 3$ either increases final reach at approximately fixed displacement or contracts both displacements in the observed score-transition region. The displayed current-state quantities exclude Polyak target lag and smoothing noise. Labels indicate T and marker size is proportional to score.

target smoothing is .9993. The close alignment is an empirical property of these checkpoints, where Polyak lag is small, rather than a mathematical equivalence. Among the 84 Prox-3/Prox-2 runs in which Prox-3 lowers deterministic target exposure, Prox-3 is nevertheless worse by more than one score point in 17. This rejects the stronger interpretation that minimizing exposure alone determines performance.

In the simulator-reference screening, 55 stable checkpoints contribute 660 states and five collapsed checkpoints contribute 60. At a_K , the median of checkpoint-level state-median signed errors relative to $Q_{\text{MC}}^{\rho_1}$ is -41.4 in stable checkpoints and $+1.70 \times 10^{12}$ in collapsed checkpoints. Pooling states instead gives a stable median of -38.7 ; the checkpoint-level aggregate is the inferential summary used here. This provides simulator-reference corroboration of the overestimation signature in Figure 3. We do not treat the sparse subset as a population action-ranking estimate: only 69 of 720 states have both action differences outside the dead zone, with 29 sign disagreements, and within-checkpoint states are correlated.

Table 5: Targeted Hopper-medium scores. The $K \leq 3$ entries use the audit-matched seed-0/1 pair; Prox-4 reports the four-seed mean $\pm$ sample standard deviation and supplies stability-window context rather than a matched movement comparison.

T	TD3+BC	Prox-2	Prox-3	Prox-4
7	48.1	83.2	94.4	84.8 $\pm$ 8.0
10	0.7	50.8	99.5	100.4 $\pm$ 1.1
12	0.9	0.7	93.6	101.2 $\pm$ 0.4
14	0.7	50.6	88.1	101.1 $\pm$ 0.7
17	0.7	0.6	0.7	100.9 $\pm$ 1.7
20	0.7	0.7	51.5	41.8 $\pm$ 44.0

Table 6: Host-separated full-grid replication. High is the $T \geq 4$ mean; Cell is the number of seed-pair-mean cells below 20 out of 63.

K	High 0-1	High 2-3	Cell 0-1	Cell 2-3
1	25.97	24.79	35	35
2	42.02	40.38	25	26
3	52.78	51.05	17	16
4	63.61	64.33	9	10

Table 7: Targeted final-return controls on the same eight environment–budget–seed cells. The three $K = 3$ rows are compute matched to three actor optimizer calls; the $K = 1$ rows differ only in actor-target Polyak rate.

Condition	Mean	Median	Score < 20
Sequential $K = 3$	71.4	88.9	2/8
Fixed-ref. $K = 3$	70.0	87.9	2/8
Direct $K = 3$	59.4	89.6	3/8
Standard $K = 1$	13.6	0.8	6/8
3 $\times$ target lag, $K = 1$	12.0	1.5	7/8

Table 8: Matched-cell audit of deterministic next-state Polyak-target exposure and current-state final-policy displacement. The Lower column counts runs in which the left method has the smaller quantity.

Comparison	Quantity	Lower	Ratio
P3 / TD3	Target disp.	88/90	.658
P3 / TD3	Final disp.	36/90	1.057
P3 / P2	Target disp.	84/90	.903